UNIT 3 · TOPIC 4(A)

Testing of Reapers, Mowers and Combines

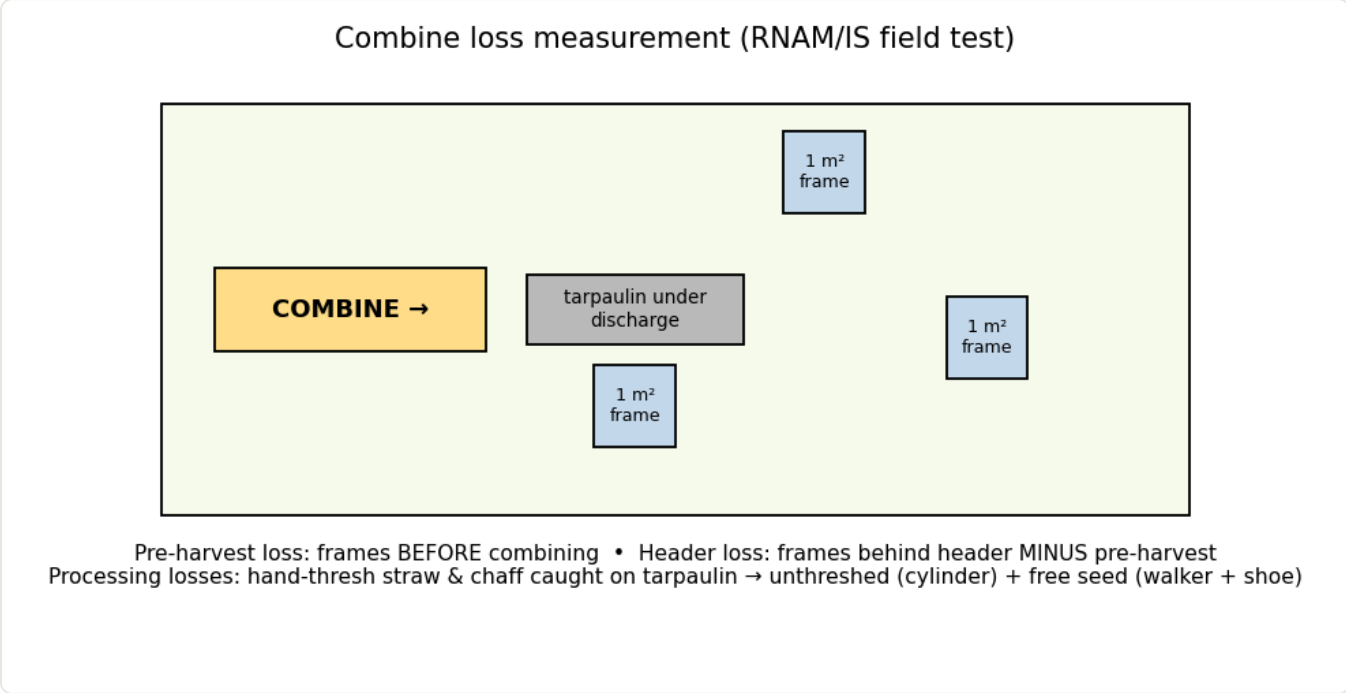
Syllabus topic: Testing of harvesting and threshing equipment, grain and straw combines

Reaper/mower field test

Measure: effective cutting width, stubble height and uniformity, UNCUT crop %, cutting/conveying loss (shatter + dropped heads), bunching quality (reaper-binders/windrowers), field capacity, fuel and power.

1. Count the grains fallen in 1-m² frames after the operation.
2. Subtract the pre-harvest shatter.
3. Convert to kg/ha using the 1000-grain weight.

Combine harvester test (RNAM/IS) — the loss chain



Combine loss measurement (RNAM/IS field test)

1. **Pre-harvest (natural) shatter:** place frames in the standing crop BEFORE the pass. Not chargeable to the machine.
2. **Header loss:** free grain + heads on the ground behind the cut, MINUS the pre-harvest loss.
3. **Processing losses:** catch the rear discharge on a tarpaulin over a measured strip. Hand-thresh the straw/chaff → CYLINDER (unthreshed) loss; free grain in straw = WALKER loss; free grain in chaff = SHOE loss.

Also measure: grain DAMAGE % (sieve/visual check of a tank sample), CLEANLINESS % of tank grain, MOG feed rate (kg/s), fuel and capacity. Loss calculations are in Topic 7b.

Practice MCQs

Q79. Pre-harvest (natural shatter) loss is measured by:

(a) Frames placed AFTER combining
(b) Frames placed in standing crop BEFORE harvest
(c) Tank sampling
(d) Weighing straw

Q80. Header loss equals grain found behind the header:

(a) Plus pre-harvest loss
(b) Minus pre-harvest loss
(c) Divided by yield
(d) Times swath

Q81. Unthreshed grain in combine discharge is charged to the:

(a) Header
(b) Cylinder
(c) Shoe
(d) Reel

Q82. Free grain found in the straw from the rear of a combine is the:

(a) Shoe loss
(b) Walker/rack loss
(c) Header loss
(d) Pre-harvest loss

Q105. Combine grain-tank sample is analysed for:

(a) Cleanliness and damage %
(b) Draft
(c) Slip
(d) MWD

Answers

Q79 — (b) Frames placed in standing crop BEFORE harvest
Q80 — (b) Minus pre-harvest loss
Q81 — (b) Cylinder
Q82 — (b) Walker/rack loss
Q105 — (a) Cleanliness and damage %

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities) & Ch. 17; RNAM test codes (combine, thresher, transplanter); BIS standards for power threshers and chaff cutters; Jagdishwar Sahay; NPTEL lectures.